INCH-POUND

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MILITARY SPECIFICATION

RANGES, PACKAGING AND PACKING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers optional detailed requirements or performance requirements for packaging and packing for shipment and storage of household, field, and heavy duty hotel-type cooking ranges, and floor-type deep fat fryers, griddles, and baking and roasting ovens, heated by electricity, gas, gasoline, oil, wood, or coal.

1.2 Classification.

1.2.1 Options. Packaging and packing requirements shall be of the following options as specified (see 6.2).

Option I - Detailed requirements

Option II - Performance requirements

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1898	-	Cushioning Material, Cellulosic, Packaging
L-P-378	-	Plastic, Sheet and Strip, Thin Gauge, Polyolefin
NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way, (Partial)
UU-P-134	-	Paper, Wrapping, Wet-Waxed
UU-P-268	-	Paper, Kraft, Wrapping
UU-T-81	-	Tags, Shipping and Stock
MMM-A-260	-	Adhesive, Water-Resistant (For Sealing Waterproofed Paper)
PPP-B-576	-	Boxes, Wood-Cleated Panelboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-1163	-	Box, Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Wax-Resin Impregnated
PPP-B-1364	-	Boxes, Shipping, Corrugated Fiberboard, High Strength, Weather Resistant, Double Wall
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electric and Mechanical - Requirements for Packaging Of
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	-	Tape; Packaging/Industrial, Filament Reinforced

MILITARY

MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation, Methods Of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible

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MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000-Pound Capacity
MIL-C-11133 - Crates, Shipping, Wood, Open, Wirebound
MIL-C-46842 - Cushioning Material, Unicellular Polyethylene Foam (For Packaging Purpose)
MIL-C-52950 - Crates, Wood, Open and Covered

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-11-795 - Ranges; Containers for Level A and Level B Shipment

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-UX, Natick, MA 01760-5017.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3950 - Standard Specification for Strapping, Plastic (and seals)
D 3951 - Standard Practice for Commercial Packaging
D 3953 - Standard Specification for Strapping, Flat Steel and Seals
D 4727 - Standard Specification for Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes

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(Application for copies shall be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Selection of options. Units having a combined gross weight (complete unit and packaging materials and shipping container) not exceeding 800 pounds shall be packaged and packed in accordance with either option I or II at the discretion of the contractor. Units with a combined gross weight exceeding 800 pounds shall be packaged in accordance with option I (see 6.2).

3.2 Packaging. Packaging shall be level A or C as specified (see 6.2).

3.2.1 Level A.

3.2.1.1 Option I. Detailed requirements.

3.2.1.1.1 Material. Material shall be as specified herein.

3.2.1.1.2 Cleaning and preservation. The cleaning process and application of and type of preservation shall be in accordance with MIL-P-116 as specified herein.

3.2.1.1.3 Disassembly. Detachable leveling pads, drip trays, racks, burner grates, broiler pans, and accessories shall be removed. Surface heating units or burners and oven and broiler heating units shall either be removed or cushioned with material specified in 3.2.1.1.11, or with corrugated fiberboard, and secured in their respective position with either clips or with wire of not less than 16 gage, or blocked or braced in such a manner as to prevent damage during shipment. Projecting parts, easily removed, may be detached from the unit for the reduction of cubage. Nuts, bolts, and screws removed during disassembly shall be reinserted and secured in their respective locations. Alternatively, the nuts, bolts, and screws may be placed in a cloth drawstring bag, the bag cushioned and securely attached in a conspicuous place on the unit in a manner so as not to damage the finish of the unit.

3.2.1.1.4 Matchmarking. Cloth shipping tags conforming to type A, size 1 of UU-T-81 shall be attached securely to the parts removed and to mating

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parts on the unit in such a manner as not to damage the item or the preservation thereof. Suitable matchmarking and identification shall be printed on these tags.

3.2.1.1.5 Cleaning. All exposed, uncoated, unpainted ferrous metal surfaces of the unit, accessories and parts shall be thoroughly cleaned by process C-1.

3.2.1.1.6 Repainting and recoating. Painted and permanently coated surfaces from which the paint and permanent coating have been removed, or which are defective, shall be cleaned, treated, and repainted or recoated in accordance with the original requirements.

3.2.1.1.7 Application of preservative. All cleaned, uncoated, and unpainted ferrous metal surfaces, except corrosion-resisting steel, shall be coated with type P-14 preservative.

3.2.1.1.8 Protection of preservative. All exposed surfaces of the units, accessories, and parts, coated with preservatives that might contact other surfaces, including packaging material, shall be covered, wrapped, or bagged with greaseproof barrier material conforming to type II, grade A, class 1 or 2 of MIL-B-121. Covering, wrapping, or bagging shall be heat sealed or secured with tape conforming to PPP-T-45 or jute, sisal, manila, or cotton twine.

3.2.1.1.9 Operating instructions. Operating instructions and service manuals shall be individually packaged in accordance with method IC-1 or method IC-3 and securely attached to the unit.

3.2.1.1.10 Accessories and parts. Component parts and accessories, disassembled as specified in 3.2.1.1.3, shall be cushioned by wrapping in cushioning materials, as specified in 3.2.1.1.11 and then packaged in a fiberboard box or boxes conforming to type CF, class domestic, variety SW, grade 200, style RSC of PPP-B-636. Each fiberboard box shall be securely closed in accordance with the appendix of the box specification. The fiberboard box or boxes shall be secured either within the unit or in an unused section of the shipping container.

3.2.1.1.11 Doors and drawers. All doors and drawers of the unit shall be cushioned in a closed position with cushioning material conforming to grade III, class B, style 1 size optional of A-A-1898 or MIL-C-46842. The cushioning material shall be placed between the body of the unit and points of contact of the doors, drawers, and other panels to prevent chipping of the porcelain enamel finish. The doors and drawers shall be secured in the closed position with steel strapping 1/4 by 0.015 inch on the household units and 3/8 by 0.020 inch on the heavy duty, or hotel-type unit. Strapping shall conform to type I, class A, of ASTM D 3953. A continuous strip of cushioning material conforming to grade III, class B, style 1 size optional of A-A-1898

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shall be placed under the steel strapping to prevent contact of the steel straps with the porcelain or enamaled surfaces of the unit. The strip of cushioning material shall be minimum 1/4 inch thick and 1-1/2 inches wide, completely engirdling the unit and shall be centered under the steel strapping. Alternatively, the doors and drawers may be secured in closed position with a minimum 1/2 inch width tape on all heavy duty or hotel-type units and floor-type fryers, baking and roasting ovens. The tape shall conform to type III of PPP-T-97 and shall show no staining when tested in accordance with 4.3.1. The tape shall be applied firmly with a hard rubber roller and shall extend not less than 6 inches on the door or drawer and onto each adjacent end or side surface no less than 6 inches. In lieu of steel strapping and tape banding, nonmetallic strapping conforming to type II or III of ASTM D 3950 may be used provided that the breaking strength is 310 pounds for household units, and 420 pounds for heavy duty or hotel-type. When nonmetallic strapping is used, no cushioning is required and seals shall be applied on the back surface of the unit.

3.2.1.2 Option II, performance requirements (see 3.1).

3.2.1.2.1 Preservation. Preservation shall be as specified in 3.2.1.1 through 3.2.1.1.8.

3.2.1.2.2 Packaging. Accessories, parts, doors, and drawers shall be cushioned and packed in a manner to meet the performance test requirements for level A packing of PPP-P-600.

3.2.2 Level C. Each unit shall be cleaned, preserved, and packaged to afford adequate protection against deterioration and damage during shipments from the supply source to the first receiving activity in accordance with ASTM D 3951. The contractor may use standard commercial practice when it meets these requirements.

3.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Option I, detailed requirements (see 3.1). Each unit, together with accessories and parts, preserved and packaged as specified in 3.2.1.1 or 3.2.2 shall be packed as indicated in table I (see 3.3.4) for weight limitations and applicable securing methods in shipping containers specified in 3.3.1.1.1, 3.3.1.1.2, or 3.3.1.1.3, as applicable.

3.3.1.1.1 Box, fiberboard, maximum weight of contents 65 pounds. The fiberboard shipping containers shall be snug-fitting conforming to style optional V2s or V3s of PPP-B-636. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells made of the type CF, variety SW, class weather-resistant, grade W5c, A or C flute of ASTM D 4727.

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Each fiberboard shipping container shall be closed in accordance with method II, waterproofed in accordance with method V and reinforced in accordance with the appendix of PPP-B-636. The maximum weight of contents of V3s shall be 40 pounds and of V2s shall be 65 pounds.

3.3.1.1.2 Boxes, wood, cleated-plywood, maximum weight of contents 1,000 pounds. Wood, cleated-plywood shipping containers shall conform to overseas type, grade A or B, of PPP-B-601, and as shown in Drawing No. 5-11-795.

3.3.1.1.3 Crates, wood; lumber and plywood sheathed, nailed and bolted, weight of contents exceeding 1,000 pounds. Sheathed nailed wood crates shall conform to type I or II, class 1 or 2, style A of MIL-C-104.

3.3.1.2 Option II, performance requirements (see 3.1). Each unit preserved and packaged as specified in 3.3.1.2 or 3.3.2 shall be packed in accordance with the level A requirements of PPP-P-600. The packed unit shall meet the level A performance requirements of PPP-P-600.

3.3.2 Level B.

3.3.2.1 Option I, detailed requirements (see 3.1). Each unit, together with accessories and parts, preserved and packaged as specified in 3.3.1.1 or 3.3.2, shall be packed as indicated in table I (see 3.3.4) for weight limitations and applicable securing methods, in shipping containers specified in 3.3.2.1.1 through 3.3.2.1.7, as applicable. The contents of each open type crate shall be protected from the elements by a full length shroud fabricated of minimum 60 pound kraft wrapping paper conforming to type I, grade A of UU-P-268. Seams shall be sealed with water-resistant adhesive conforming to MMM-A-260. Alternatively, the shroud shall be fabricated from nominal 3 mil polyethylene conforming to L-P-378. All seams of the polyethylene shroud shall be heat sealed. The shroud shall hang free, when possible, and shall extend to the floor of the crate base. All sharp points shall be cushioned to prevent rupture or chafing of the shroud. All open type crates shall be assembled, closed, and strapped in accordance with the appendix of the applicable crate specification.

3.3.2.1.1 Box, fiberboard, maximum weight of contents 120 pounds. The fiberboard shipping containers shall be snug-fitting conforming to type CF or SF, class domestic, style optional of PPP-B-636. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells made of type CF, variety SW or DW, class domestic, minimum grade 200 of ASTM D 4727. Each fiberboard shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. When specified (see 6.2), the fiberboard shipping container shall be grade V3c, V3s or V4s of PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636. The weight of contents for V3c, V3s and V4s shall not exceed 65 pounds. The weight of contents for class domestic boxes shall not exceed that specified in PPP-B-636.

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3.3.2.1.2 Boxes, fiberboard, corrugated, triple-wall, maximum weight of contents 275 pounds. Triple-wall corrugated fiberboard boxes shall conform to class 1, grade A, style E, of PPP-B-640. The contents shall be immobilized by the use of die-cut or built-up fiberboard pads or cells made of material specified in 3.3.2.1.1, except that the minimum grade shall be 275. Each fiberboard shipping container shall be securely closed in accordance with the appendix of the container specification. When specified (see 6.2), the triple-wall corrugated shipping containers shall be fabricated of class 2, weather-resistant material. When weights of contents and containers exceed 250 pounds, the container shall be strapped to a pallet conforming to NN-P-71 of size to fit the base of the container. Strapping used shall be in accordance with the appendix of PPP-B-640. Alternatively, double-wall fiberboard boxes conforming to type DW, grade 600 of PPP-B-1163 or PPP-B-1364 may be used.

3.3.2.1.3 Boxes, wood-cleated, panelboard, maximum weight of contents 400 pounds. Wood-cleated panelboard containers shall conform to type I or III, class 1 of PPP-B-576 and as shown in Drawing No. 5-11-795.

3.3.2.1.4 Boxes, wood, cleated-plywood, maximum weight of contents 1,000 pounds. Wood, cleated-plywood shipping containers shall conform to domestic type, grade A or B of PPP-B-601 and as shown on Drawing No. 5-11-795.

3.3.2.1.5 Crates, shipping, wood, open, wirebound, maximum weight of contents 1,000 pounds. Wirebound shipping crates shall conform to type II, class 1 or 2, style 1 or 2, grade B; or type III, class 1, 2, or 3, style 1 or 2, grade B of MIL-C-11133, as applicable.

3.3.2.1.6 Crates, wood, open and covered, maximum weight of contents 2,500 pounds. Open wood crates shall conform to style A, type I, II, IV, or V, class 1 of MIL-L-52950, as applicable.

3.3.2.1.7 Crates, wood; open, 12,000- and 16,000-pound capacity, contents exceeding 2,500 pounds. Open wood crates shall conform to type I, style A, of MIL-C-3774.

3.3.2.2 Option II, performance requirements (see 3.1). Each unit, preserved and packaged as specified in 3.3.1.2 or 3.3.2, shall be packed in accordance with the level B requirements of PPP-P-600. The packed unit shall meet the level B performance requirements of PPP-P-600.

3.3.3 Level C. Units, together with accessories and parts, preserved and packaged as specified in 3.2.2, shall be secured and packed in shipping containers in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies in accordance with ASTM D 3951.

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3.3.4 Securing of units in the shipping container. Each unit shall be secured within the shipping container by the methods as indicated in table I for the weight limitations specified. A minimum of 1 inch clearance shall be provided between the top, sides and ends of the unit and the adjacent inside surfaces of the shipping container.

TABLE I. Methods of securing units in shipping containers and maximum weight of contents

Container	Securing method	Weight of contents (pounds)
Boxes, Shipping, Fiberboard, PPP-B-636	See 3.3.1.1.1 and 3.3.2.1.1	Unless otherwise specified the weight of contents shall not exceed the normal-weight limitation of the specification.
Boxes, Fiberboard, Corrugated PPP-B-640, PPP-B-1163, or PPP-B-1364	See 3.3.2.1.2	275 maximum
Boxes, Wood-Cleated Panelboard a/ Type I or III of PPP-B-576 and Drawing No. 5-11-795	2 1 or 3 or 1 and 3	300 maximum 400 maximum
Boxes, Wood, Cleated-Plywood, PPP-B-601 and Drawing No. 5-11-795	2 b/ 1 or 3 b/ or 1 and 3 b/	300 maximum 1,000 maximum
Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted, MIL-C-104	1 or 3 b/ or 1 and 3 b/	Exceeding 1,000
Crates, Shipping, Wood, Open, Wirebound, MIL-C-11133	2 1 or 3 or 1 and 3	300 maximum 1,000 maximum
Crates, Wood, Open and Covered, MIL-C-52950	2 1 or 3 or 1 and 3	300 maximum 2,500 maximum
Crates, Wood; Open 12,000- and 16,000-Pound Capacity, MIL-C-3774	1 or 3 or 1 and 3	Exceeding 2,500

a/ For level B shipments only.

b/ Cushioning material shall be water-resistant for level A shipment.

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3.3.4.1 Method 1 (bolting). Units, fabricated with provision for bolting, shall be secured to the base of the container as indicated in 3.3.4.1.1.

3.3.4.1.1 Base. The unit shall be shipped upright and secured to the base of the shipping container with carriage or machine bolts extending through the load bearing members of the base or through the skids and the load bearing members of the base into the unit base or legs. The heads of the bolts shall be at the underside of the load bearing members or the skids, and not less than four bolts shall be used for securing the unit to the crate base. Carriage bolt diameters shall be minimum 5/16 of an inch for units weighing less than 400 pounds, and 3/8 of an inch for units weighing not more than 1,500 pounds. Units in excess of 1,500 pounds shall require 1/2 inch diameter carriage bolts. Round flat washers shall be used under the heads of all machine bolts, and lockwashers shall be used under all nuts.

3.3.4.2 Method 2, pressure pack. The unit shall be cushioned and held securely within the shipping container by built-up single-wall, double-wall or triple-wall corrugated fiberboard edge pads or cushioning material conforming to MIL-C-46842 of adequate thickness to provide a minimum of 1 inch clearance between any protruding part of the unit and the interior surface of the shipping container, and shall be of adequate width to provide a minimum width contact area of 3 inches on all horizontal edges at top, bottom, and sides, and vertical end edges of the unit. When the splash back is an integral part of the unit, a sheet of double-walled corrugated fiberboard conforming to ASTM D 4727 or cushioning material conforming to MIL-C-46842 having a minimum of 1/4 inch thickness shall completely cover the top of the unit and shall have a top locking (hold-down) consisting of half-slotted partitions, spacing not exceeding 4 inches, fabricated of 275 pound test double-walled corrugated fiberboard conforming to ASTM D 4727 placed on top of the pad covering the top of the unit. The partitions shall be of adequate height to provide a minimum of 1 inch clearance above the top of the splash back. In lieu of the pad and half-slotted top blocking (hold-down), any similar corrugated fiberboard blocking assembly or cushioning may be used provided it protects the top of the unit and allows a minimum of 1 inch clearance above the splash back. Method 2 shall not be used for units weighing more than 300 pounds. At points of contact with porcelain enamel surfaces, corrugated fiberboard shall be provided with nonabrasive wax coating (see 3.3.5.c).

3.3.4.3 Method 3, blocking and bracing. The unit shall be secured within the container by means of wood blocking and bracing. The blocking and bracing shall be cushioned at all points of contact with the unit with approximately 1/4 inch thick cellulose wadding or foam conforming to grade III, class B, style optional and size 1 of A-A-1898 or MIL-C-46842, respectively. Blocking and bracing shall not have any bearing on any surface of the unit, subject to damage by distortion or abrasion. The blocking and bracing shall not depend on end grain nailing to the container, but shall be supplemented with supporting blocks and cleats.

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3.3.5 Protection of finish, enamel or porcelain surfaces for method 2 or 3. One of the following shall be placed between the cushioned blocking or corrugated fiberboard pads and the finished surface or enamel or porcelain surfaces of the unit:

(a) A sheet of wax paper conforming to type III of UU-P-134.
(b) Cellulosic cushioning material having a thickness of not less than 0.20 inch conforming to grade III, class B, style 1 size K of A-A-1898.
(c) Corrugated fiberboard with nonabrasive wax coating.
(d) A sheet of nominal 3 mil polyethylene film conforming to L-P-378.

3.4 Marking.

3.4.1 Interior packages. Interior packages shall be marked in accordance with MIL-STD-129. Component parts that have been overwrapped and secured in their respective locations will not require marking.

3.4.2 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

3.4.2.1 Precautionary marking. Special handling marking requirements in MIL-STD-129 applicable to fragile items, such as arrows, the words "UP" and "THIS SIDE UP", and "CENTER OF BALANCE" shall apply.

3.5 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document.

3.6 Workmanship. The end item shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification

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shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1. Tape. Tape shall be tested in accordance with 4.3.1 through 4.3.1.3, to determine compliance with 3.2.1.1.11. Any evidence of staining shall be cause for rejection of the tape.

4.2.2 Examination of the end item. The end items shall be examined for visual preparation for packaging, preservation and packaging, packing, and marking in accordance with the defects listed in tables II through V. The lot size shall be expressed in units of disassembled, packaged or packed ranges, as applicable. The sample unit shall be one disassembled, packaged or packed range, of the specified type, design and characteristic. The inspection levels and the acceptable quality levels (AQLs), expressed in terms of defects per hundred units, shall be as follows:

Table No.	Inspection levels	AQLs Major	AQLs Total 1/
II	S-2	2.5	10.0
III	S-2	2.5	
IV	S-2	2.5	
V	S-2	2.5	

1/ Total represents major and minor defects combined for the AQL value.

TABLE II. End item visual preparation for packaging defects

Examine	Defect	Category Major	Category Minor
Disassembly	Detachable parts and accessories not removed (when specified)	101	

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TABLE II. End item visual preparation for packaging defects (cont'd)

Examine	Defect	Category Major	Category Minor
Disassembly (cont'd)	Detachable parts and accessories not cushioned and secured in position or bagged and cushioned (when specified)	102	
	When nuts, bolts, and screws are bagged the bag is not cushioned, or secured in a conspicuous place	103	
Matchmarking	Tags with information not attached to mating parts		201
Repainting and recoating	Not refinished or recoated as needed	104	
	Material not identical to original	105	
Preservative coating	Barrier missing or not covering coating at contact points	106	
	Barrier not taped or tied in position	107	

TABLE III. End item for preservation and packaging defects

Examine	Defect	Category Major	Category Minor
Option I - Levels A and C			
Operating instructions and service manual	Not placed individually in envelope		201
	Not securely attached to unit		202
	Wrong method of packaging	101	
Door and drawers	Not secured as specified	102	
	Incorrect width and thickness of cushioning	103	
	Not cushioned as specified	104	
	Cushioning material missing (when specified)	105	
	Incorrect size of strap	106	
	Strap not centered on cushioning material when required		203
	Strap tensioned too tightly causing damage	107	

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TABLE III. End item for preservation and packaging defects (cont'd)

Examine	Defect	Category Major	Minor
Door and drawers (cont'd)	Strap tensioned too loosely causing slippage	108	
	Not taped as specified (when required)	109	
Accessories and parts (boxed)	Not wrapped with cushioning material	110	
	Not packaged in box specified	111	

TABLE IV. End item packing defects

Examine	Defect	Category Major	Minor
Option I - Levels A, B, and C			
Range	Not secured and cushioned within container as specified (see 3.3.1.1.1, 3.3.2.1.2, 3.3.4, 3.3.4.1, 3.3.4.2, and 3.3.4.3)	101	
	Finished surfaces not protected from abrasion with material specified (see 3.3.5)	102	
Accessories and parts box	Not securely blocked and braced within range (when specified)	103	
	Not secured within the shipping container (when specified)	104	
Container	Not as specified for weight limits of range (see table I)	105	
	Less than 1 inch clearance specified between range and inside of container	106	
Shroud	Shroud missing (when specified)	107	
Bolts and nuts	Less than four bolts holding range when required	108	
	Diameter of bolt less than specified when required	109	
	One or more flat washers missing under heads of machine bolts when required	110	

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TABLE IV. End item packing defects (cont'd)

Examine	Defect	Category Major	Minor
Option I - Levels A, B, and C (cont'd)			
Bolts and nuts (cont'd)	One or more nuts loose on bolts	111	
	One or more lockwashers missing	112	
	Bolts not positioned through container member and blocking as specified	113	
Blocking members	Incorrectly placed	114	
Edge pad	Not built-up of material specified	115	
	Less than length required to cover edge	116	
	Less than three inches of pad contacting each adjacent corner surface of range	117	
	Does not provide minimum one inch clearance beyond outmost projection	118	
	Not fitting as required to make tight pack	119	
Tray	Not double faced corrugated when required	120	
	Less than 4 inches deep	121	
	Missing on top	122	

TABLE V. End item marking defects

Examine	Defect	Category Major	Minor
Marking Package and shipping container (as applicable)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.	101	

4.2.3 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, PPP-B-640, and MIL-C-104, examination for defects in closure and strapping shall be in accordance with the applicable container specification.

4.2.3.1 Examination of shipping containers when PPP-B-601 is required. When shipping containers are required to be in accordance with PPP-B-601, examination of defects in closure and strapping for determination of

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compliance with the specification, as applicable, and with Drawing No. 5-11-795 shall be in accordance with the defects listed in table VI. The lot size shall be expressed in shipping containers. The sample unit shall be one shipping container. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for total (major and minor combined) defects.

TABLE VI. Strapping and closure defects

Examine	Defect	Category Major	Category Minor
Strapping	Not applied as specified (see Drawing No. 5-11-795)	101	
	Missing strap	102	
	Loose strap	103	
	Torn strap	104	
	Cut strap	105	
	Not drawn tight	106	
	Weak joint	107	
Staples	Staples less than 4 inches from edge of member		201
	Less than size specified	108	
	Not spaced as specified		202

4.3 Test methods.

4.3.1 Staining test. The staining test shall be performed as specified in 4.3.1.1., 4.3.1.2, and 4.3.1.3.

4.3.1.1 Conditioning. The tape shall be conditioned for testing for not less than 24 hours at 73 ± 2°F, and 50 percent ± 4 percent relative humidity.

4.3.1.2 Specimens. The first three layers of tape shall be removed from the rolls before taking specimens to be tested. An equal number of 10 inch lengths of tape shall be taken as specimens from all sample rolls to be tested.

4.3.1.3 Staining. Specimens of tape shall be applied adhesive down to a clean porcelain enamel surface panel and rolled down with two passes, once in each longitudinal direction, using a 4-1/2 pound rubber roller having a surface durometer hardness between 70 and 80 and approximately 3-1/2 inches in diameter, at a rate of travel of 10 inches per minute. Panels shall be placed in an oven having a temperature of 150 ± 2°F for a period of 10 days. Upon removal, panels shall be allowed to cool 2 hours before stripping

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off the tape. Tape edge transfer may be noticed due to the high temperature. If so, it should be removed with solvent prior to determination for staining. Tape shall be rejected for use if staining is evident.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The packaging and packing of ranges and related items.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Options required (see 3.1).
d. Levels of packaging and packing required (see 3.2 and 3.3).
e. Whether class weather-resistant material is used for level B packing (see 3.3.2.1.1 and 3.3.2.1.2).

6.3 Subject term (keyword) listing.

Detailed requirements
Performance requirements
Preservation
Options

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - YD
Air Force - 99

Review activities:

Army - MD, SM
Navy - SA
Air Force - 69, 84
DLA - GS

User activity:

Navy - MC

Preparing activity:

Army - GL

(Project 7310-0813)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-11337G	1992 December 31

3. DOCUMENT TITLE

RANGES, PACKAGING AND PACKING OF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON/DSN 256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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